Software Engineering for CS

Week 6

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Switch off mobile phones during lectures, or put them into silent mode

Prototyping

- It is the technique of constructing a partial implementation of a system so that customers, users, or developers can learn more about a problem or a solution to that problem

Prototype - 1

- An initial version of the system under development, which is available early in the development process
- A prototype can be a subset of a system, and vice versa, but they are not the same

Prototype - 2

- In hardware systems, prototypes are often developed to test and experiment with system designs
- In software systems, prototypes are more often used to help elicit and validate the system requirements. There are other uses also

Prototype - 3

- It should be easy for a prototype to be developed quickly, so that it can be used during the development process
- Prototypes are valuable for requirements elicitation because users can experiment with the system and point out its strengths and weaknesses. They have something concrete to criticize

Types of Prototyping

- Throw-away prototyping
- Evolutionary prototyping

Throw-away Prototyping - 1

- Intended to help elicit and develop the system requirements
- The requirements which should be prototyped are those which cause most difficulties to customers and which are the hardest to understand. Little documentation is needed

Throw-away Prototyping - 2

- Determine the feasibility of a requirement
- Validate that a particular function is really necessary
- Uncover missing requirements
- Determine the viability of a user interface

Throw-away Prototyping - 3

- Writing a preliminary requirements document
- Implementing the prototype based on those requirements
- Achieving user experience with prototype

Throw-away Prototyping - 4

- Writing the real SRS
- Developing the real product

Evolutionary Prototyping - 1

- Intended to deliver a workable system quickly to the customer
- The requirements which should be supported by the initial versions of this prototype are those which are well-understood and which can deliver useful end-user functionality

Evolutionary Prototyping - 2

- Documentation of the prototype is needed to build upon
- This process repeats indefinitely until the prototype system satisfies all needs and has thus evolved into the real system

Evolutionary Prototyping - 3

- Evolutionary prototype may not be built in a ‘dirty’ fashion. The evolutionary prototype evolves into the final product, and thus it must exhibit all the quality attributes of the final product

Comparison of Prototyping - 1

	Throwaway	Evolutionary
Development approach	Quick and dirty. No rigor	No sloppiness. Rigorous

Comparison of Prototyping - 2

	Throwaway	Evolutionary
What to build	Build only difficult parts	Build understood parts first. Build on solid foundation

Comparison of Prototyping - 3

	Throwaway	Evolutionary
Design drivers	Optimize development time	Optimize modifiability
Ultimate goal	Throw it away	Evolve it

Prototyping Benefits - 1

- The prototype allows users to experiment and discover what they really need to support their work
- Establishes feasibility and usefulness before high development costs are incurred

Prototyping Benefits - 2

- Essential for developing the 'look and feel' of a user interface. Helps customers in 'visualizing' their requirements
- Forces a detailed study of the requirements which reveals inconsistencies and omissions

Prototyping Costs

- Training costs
  - Prototype development may require the use of special purpose tools
- Development costs
  - Depend on the type of prototype being developed

Prototyping Problems - 1

- Extended development schedules
  - Developing a prototype may extend the schedule although the prototyping time may be recovered because rework is avoided

Prototyping Problems - 2

- Incompleteness
  - It may not be possible to prototype emergent system requirements

Additional Benefits of Prototyping

- Developing a system prototype is worth the investment in time and money
- Real needs of the customers will be reflected in the requirements set
- Rework will be reduced
- Defect prevention

Developing Prototypes

- Conventional system development techniques usually take too long, and prototypes are needed early in the elicitation process to be useful
- Rapid development approaches are used for prototype development

Approaches to Prototyping

- Paper prototyping
- ‘Wizard of Oz’ prototyping
- Executable prototyping

Paper Prototyping - 1

- A paper mock-up of the system is developed and used for system experiments
- This is very cheap and very effective approach to prototype development
- No executable software is needed

Paper Prototyping - 2

- Paper versions of the screens, which might be presented to the user are drawn and various usage scenarios are planned
- For interactive systems, this is very effective way to find users' reactions and the required information

Comments on Prototyping

- Prototyping interactive applications is easier than prototyping real-time applications
- Prototyping is used better to understand and discover functional requirements, as compared to non-functional requirements

Thanks